Title: HW6 - Machine Learning on Cloud SQL Data

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GitHub link: https://github.com/aumghelani/AUM-U07079548-CLOUD-COMPUTING-HW/tree/main/hw6

Brief: This assignment builds on HW5. The hw5-mysql Cloud SQL database is already populated with ~100,000 requests from the HW5 client runs."

Step 1) - 3NF Schema

Old schema (HW5):
requests(id PK, created_at, country, client_ip, gender, age, income, is_banned, time_of_day, requested_file)
errors(id PK, created_at, requested_file, error_code)
Problem: Transitive dependency: id → client_ip → country. Each IP always maps to the same country, but country is stored redundantly in every row. This violates 3NF.

New schema (3NF):
requests(id PK, created_at, client_ip, gender, age, income, is_banned, time_of_day, requested_file)
ip_country(client_ip PK, country)
errors(id PK, created_at, requested_file, error_code)
Migration queries:

```
-- 1. Create lookup table
CREATE TABLE ip_country (
   client_ip VARCHAR(45) NOT NULL,
   country   VARCHAR(100) NOT NULL DEFAULT '',
   PRIMARY KEY (client_ip)
);

-- 2. Populate from existing data
INSERT IGNORE INTO ip_country (client_ip, country)
SELECT DISTINCT client_ip, country FROM requests;

-- 3. Remove redundant column
ALTER TABLE requests DROP COLUMN country;
```

Why this is 3NF: Every non-key column now depends only on the primary key. Country is retrieved via JOIN on client_ip

```
[8/9] ============================================================
      MODEL RESULTS FROM GCS
      ============================================================

--- 3NF Normalization Output ---
============================================================
3NF SCHEMA MIGRATION
============================================================

Transitive dependency identified:
  id -> client_ip -> country
  (Each IP always maps to exactly one country)

Migration steps:
  1. Create ip_country(client_ip PK, country) lookup table
  2. Populate ip_country from existing requests data
  3. Drop 'country' column from requests table

[normalize] Running step 1/3 ...
  SQL: CREATE TABLE IF NOT EXISTS ip_country (
       client_ip   VARCHAR(45)   NOT NULL,
       country     VARCHAR(100)  NO...
  Done. Rows affected: 0

[normalize] Running step 2/3 ...
  SQL: INSERT IGNORE INTO ip_country (client_ip, country)
    SELECT DISTINCT client_ip, country
    FROM requests;...
  Done. Rows affected: 43586

[normalize] Running step 3/3 ...
  SQL: ALTER TABLE requests DROP COLUMN country;...
  Done. Rows affected: 0

============================================================
MIGRATION COMPLETE
  ip_country table : 43586 unique IP->country mappings
  requests table   : 100170 rows (country column removed)

New schema (3NF):
  requests(id PK, created_at, client_ip, gender, age, income,
           is_banned, time_of_day, requested_file)
  ip_country(client_ip PK, country)
  errors(id PK, created_at, requested_file, error_code)
```

```
To get country for a request, JOIN:
  SELECT r.*, ic.country
  FROM requests r
  JOIN ip_country ic ON r.client_ip = ic.client_ip;
============================================================

--- Model 1 Results (IP -> Country) ---
MODEL 1: Predict Country from Client IP
Algorithm: DecisionTreeClassifier
Features: IP address octets (4 integers)
Total samples: 100166, Train: 80132, Test: 20034
Test Accuracy: 100.00%
Training Time: 0.54s

Classification Report:
                                   precision    recall  f1-score   support

                     Afghanistan       1.00      1.00      1.00        87
                         Albania       1.00      1.00      1.00       121
                         Algeria       1.00      1.00      1.00       105
                         Andorra       1.00      1.00      1.00       104
                          Angola       1.00      1.00      1.00       112
             Antigua and Barbuda       1.00      1.00      1.00       109
                       Argentina       1.00      1.00      1.00        91
                         Armenia       1.00      1.00      1.00       125
                       Australia       1.00      1.00      1.00        85
                         Austria       1.00      1.00      1.00       108
                      Azerbaijan       1.00      1.00      1.00       121
                         Bahamas       1.00      1.00      1.00        85
                         Bahrain       1.00      1.00      1.00       106
                      Bangladesh       1.00      1.00      1.00        89
                        Barbados       1.00      1.00      1.00       102
                         Belarus       1.00      1.00      1.00        91
                         Belgium       1.00      1.00      1.00        93
                          Belize       1.00      1.00      1.00       120
                           Benin       1.00      1.00      1.00       116
                          Bhutan       1.00      1.00      1.00        87
                         Bolivia       1.00      1.00      1.00        92
           Bosnia and Herzegovina       1.00      1.00      1.00       103
                        Botswana       1.00      1.00      1.00        84
                          Brazil       1.00      1.00      1.00        81
                          Brunei       1.00      1.00      1.00        99
                        Bulgaria       1.00      1.00      1.00       107
                    Burkina Faso       1.00      1.00      1.00       109
                         Burundi       1.00      1.00      1.00        95
                      Cabo Verde       1.00      1.00      1.00        98
```

AUM – U07079548 – HW6 – CLOUD COMPUTING

Cabo Verde	1.00	1.00	1.00	98
Cambodia	1.00	1.00	1.00	109
Cameroon	1.00	1.00	1.00	109
Canada	1.00	1.00	1.00	97
Central African Republic	1.00	1.00	1.00	124
Chad	1.00	1.00	1.00	100
Chile	1.00	1.00	1.00	104
China	1.00	1.00	1.00	97
Colombia	1.00	1.00	1.00	92
Comoros	1.00	1.00	1.00	108
Congo, Democratic Republic of the	1.00	1.00	1.00	105
Congo, Republic of the	1.00	1.00	1.00	100
Costa Rica	1.00	1.00	1.00	119
Cote dIvoire	1.00	1.00	1.00	100
Croatia	1.00	1.00	1.00	104
Cuba	1.00	1.00	1.00	108
Cyprus	1.00	1.00	1.00	89
Czechia	1.00	1.00	1.00	82
Denmark	1.00	1.00	1.00	110
Djibouti	1.00	1.00	1.00	94
Dominica	1.00	1.00	1.00	114
Dominican Republic	1.00	1.00	1.00	113
Ecuador	1.00	1.00	1.00	93
Egypt	1.00	1.00	1.00	109
El Salvador	1.00	1.00	1.00	118
Equatorial Guinea	1.00	1.00	1.00	110
Eritrea	1.00	1.00	1.00	98
Estonia	1.00	1.00	1.00	102
Eswatini	1.00	1.00	1.00	106
Ethiopia	1.00	1.00	1.00	89
Fiji	1.00	1.00	1.00	102
Finland	1.00	1.00	1.00	101
France	1.00	1.00	1.00	94
Gabon	1.00	1.00	1.00	109
Gambia	1.00	1.00	1.00	105
Georgia	1.00	1.00	1.00	97
Germany	1.00	1.00	1.00	103
Ghana	1.00	1.00	1.00	80
Greece	1.00	1.00	1.00	132
Grenada	1.00	1.00	1.00	85
Guatemala	1.00	1.00	1.00	114
Guinea	1.00	1.00	1.00	87
Guinea-Bissau	1.00	1.00	1.00	82
Guyana	1.00	1.00	1.00	108
Haiti	1.00	1.00	1.00	95
Honduras	1.00	1.00	1.00	96
Hungary	1.00	1.00	1.00	90
Iceland	1.00	1.00	1.00	108

India	1.00	1.00	1.00	110
Indonesia	1.00	1.00	1.00	112
Iran	1.00	1.00	1.00	114
Iraq	1.00	1.00	1.00	91
Ireland	1.00	1.00	1.00	101
Israel	1.00	1.00	1.00	98
Italy	1.00	1.00	1.00	120
Jamaica	1.00	1.00	1.00	106
Japan	1.00	1.00	1.00	94
Jordan	1.00	1.00	1.00	111
Kazakhstan	1.00	1.00	1.00	101
Kenya	1.00	1.00	1.00	93
Kiribati	1.00	1.00	1.00	93
Kosovo	1.00	1.00	1.00	99
Kuwait	1.00	1.00	1.00	93
Kyrgyzstan	1.00	1.00	1.00	108
Laos	1.00	1.00	1.00	95
Latvia	1.00	1.00	1.00	91
Lebanon	1.00	1.00	1.00	108
Lesotho	1.00	1.00	1.00	139
Liberia	1.00	1.00	1.00	100
Libya	1.00	1.00	1.00	103
Liechtenstein	1.00	1.00	1.00	118
Lithuania	1.00	1.00	1.00	105
Luxembourg	1.00	1.00	1.00	92
Madagascar	1.00	1.00	1.00	107
Malawi	1.00	1.00	1.00	96
Malaysia	1.00	1.00	1.00	90
Maldives	1.00	1.00	1.00	104
Mali	1.00	1.00	1.00	99
Malta	1.00	1.00	1.00	90
Marshall Islands	1.00	1.00	1.00	104
Mauritania	1.00	1.00	1.00	93
Mauritius	1.00	1.00	1.00	110
Mexico	1.00	1.00	1.00	95
Micronesia	1.00	1.00	1.00	126
Moldova	1.00	1.00	1.00	105
Monaco	1.00	1.00	1.00	87
Mongolia	1.00	1.00	1.00	87
Montenegro	1.00	1.00	1.00	105
Morocco	1.00	1.00	1.00	99
Mozambique	1.00	1.00	1.00	87
Myanmar	1.00	1.00	1.00	107
Namibia	1.00	1.00	1.00	97
Nauru	1.00	1.00	1.00	102
Nepal	1.00	1.00	1.00	108
Netherlands	1.00	1.00	1.00	90
New Zealand	1.00	1.00	1.00	102

Test Set Predictions (first 100 rows):

IP Octets	Actual	Predicted	Correct
12.244.5.144	Nauru	Nauru	YES
125.99.197.218	Colombia	Colombia	YES
186.78.155.161	Cambodia	Cambodia	YES
239.13.104.164	Ethiopia	Ethiopia	YES
151.201.153.119	Croatia	Croatia	YES
27.227.37.172	Sao Tome and Principe	Sao Tome and Principe	YES
86.202.139.57	South Korea	South Korea	YES
163.78.21.179	Moldova	Moldova	YES
169.69.7.178	Mozambique	Mozambique	YES
191.140.211.92	Iceland	Iceland	YES
61.20.169.228	Kiribati	Kiribati	YES
239.169.71.167	Saudi Arabia	Saudi Arabia	YES
63.245.117.136	Cote dIvoire	Cote dIvoire	YES
194.220.43.221	Nepal	Nepal	YES
5.0.73.1	Romania	Romania	YES
83.162.52.249	Central African Republic	Central African Republic	YES
191.140.211.114	Iceland	Iceland	YES
188.31.113.239	Belgium	Belgium	YES
160.79.138.71	Saint Kitts and Nevis	Saint Kitts and Nevis	YES
186.78.155.97	Cambodia	Cambodia	YES
80.64.48.161	Sao Tome and Principe	Sao Tome and Principe	YES
113.14.123.101	Haiti	Haiti	YES
234.71.196.100	Senegal	Senegal	YES
104.45.155.169	Georgia	Georgia	YES
109.10.19.149	Argentina	Argentina	YES
185.87.35.15	Morocco	Morocco	YES
215.85.131.190	Jamaica	Jamaica	YES
28.49.36.241	Ecuador	Ecuador	YES
195.156.238.141	Suriname	Suriname	YES
86.166.107.241	Nigeria	Nigeria	YES
182.172.170.20	Togo	Togo	YES
102.240.230.44	Costa Rica	Costa Rica	YES
24.2.67.4	Sweden	Sweden	YES
104.45.155.140	Georgia	Georgia	YES
222.18.54.254	Jordan	Jordan	YES
71.183.184.53	Bangladesh	Bangladesh	YES
65.36.80.35	Turkmenistan	Turkmenistan	YES
75.185.66.234	Cabo Verde	Cabo Verde	YES
89.148.148.59	Singapore	Singapore	YES
243.212.189.154	Italy	Italy	YES
157.120.142.59	South Korea	South Korea	YES
136.213.144.190	Central African Republic	Central African Republic	YES

```
76.183.193.237       United States of America (USA) United States of America (USA) YES
149.111.3.233        France                         France                         YES
--- Model 2 Results (Fields -> Income) ---
MODEL 2: Predict Income from Request Fields
Algorithm: RandomForestClassifier (100 trees, max_depth=20)
Features: gender, age, time_of_day, country, is_banned
Total samples: 100166, Train: 80132, Test: 20034
Test Accuracy: 64.43%
Training Time: 6.81s

Feature Importances:
  country        : 0.4466
  age            : 0.3802
  time_of_day    : 0.1602
  gender         : 0.0094
  is_banned      : 0.0037

Classification Report:
              precision    recall  f1-score   support

       0-10k       0.65      0.66      0.65      2488
   100k-150k       0.64      0.65      0.65      2485
     10k-20k       0.65      0.64      0.65      2566
   150k-250k       0.64      0.63      0.63      2472
     20k-40k       0.65      0.65      0.65      2515
       250k+       0.64      0.65      0.64      2504
     40k-60k       0.63      0.63      0.63      2401
    60k-100k       0.66      0.64      0.65      2603

    accuracy                           0.64     20034
   macro avg       0.64      0.64      0.64     20034
weighted avg       0.64      0.64      0.64     20034


Test Set Predictions (first 100 rows):
Actual               Predicted            Correct
------------------------------------------------------------
100k-150k            100k-150k            YES
20k-40k              40k-60k              NO
150k-250k            60k-100k             NO
100k-150k            40k-60k              NO
150k-250k            100k-150k            NO
40k-60k              40k-60k              YES
10k-20k              10k-20k              YES
10k-20k              0-10k                NO
20k-40k              20k-40k              YES
0-10k                0-10k                YES
150k-250k            100k-150k            NO
100k-150k            100k-150k            YES
```

```
250k+                150k-250k            NO
20k-40k              20k-40k              YES
40k-60k              40k-60k              YES
40k-60k              20k-40k              NO
60k-100k             20k-40k              NO
10k-20k              10k-20k              YES
40k-60k              40k-60k              YES
0-10k                10k-20k              NO
150k-250k            150k-250k            YES
0-10k                10k-20k              NO
60k-100k             150k-250k            NO
250k+                250k+                YES
--- Models Console Output ---
HW6 - Machine Learning Models
  DB: 127.0.0.1:3306/hw5db
  Bucket: aum-hw2-u07079548

Fetching data from database ...
  Fetched 100166 rows.

============================================================
MODEL 1: Predict Country from Client IP
============================================================
  Total samples  : 100166
  Unique IPs     : 43584
  Unique countries: 197
  Train size     : 80132
  Test size      : 20034
  Training time  : 0.54s
  Test accuracy  : 100.00%

  Results written to /tmp/hw6_output/model1_results.txt

============================================================
MODEL 2: Predict Income from Request Fields
============================================================
  Samples with income: 100166
  Features       : gender, age, time_of_day, country, is_banned
  Target         : income
  Unique incomes : 8
  Income classes : [np.str_('0-10k'), np.str_('100k-150k'), np.str_('10k-20k'), np.str_('150k-250k'), np.str_('20k-40k
'), np.str_('250k+'), np.str_('40k-60k'), np.str_('60k-100k')]
  Train size     : 80132
  Test size      : 20034
  Training time  : 6.81s
  Test accuracy  : 64.43%

  Results written to /tmp/hw6_output/model2_results.txt
```

```
============================================================
UPLOADING RESULTS TO GCS
============================================================
  Uploaded: gs://aum-hw2-u07079548/hw6/model1_results.txt
  Uploaded: gs://aum-hw2-u07079548/hw6/model2_results.txt

============================================================
SUMMARY
============================================================
  Model 1 (IP -> Country)    : 100.00% accuracy
  Model 2 (Fields -> Income) : 64.43% accuracy
  Results uploaded to: gs://aum-hw2-u07079548/hw6/
============================================================
```

Step 2) - Model 1: IP → Country

How it works:

- Algorithm: DecisionTreeClassifier (scikit-learn)
- Features: 4 integer octets from the IP address (e.g., 192.168.1.1 → [192, 168, 1, 1])
- Target: country name
- Train/test split: 80/20
- Since each IP maps to exactly one country, the decision tree learns this mapping perfectly, achieving ~100% accuracy.

```
============================================================
MODEL 1: Predict Country from Client IP
============================================================
  Total samples   : 100166
  Unique IPs      : 43584
  Unique countries: 197
  Train size      : 80132
  Test size       : 20034
  Training time   : 0.54s
  Test accuracy   : 100.00%

  Results written to /tmp/hw6_output/model1_results.txt
```

Step 3) - Model 2: Fields → Income

How it works:

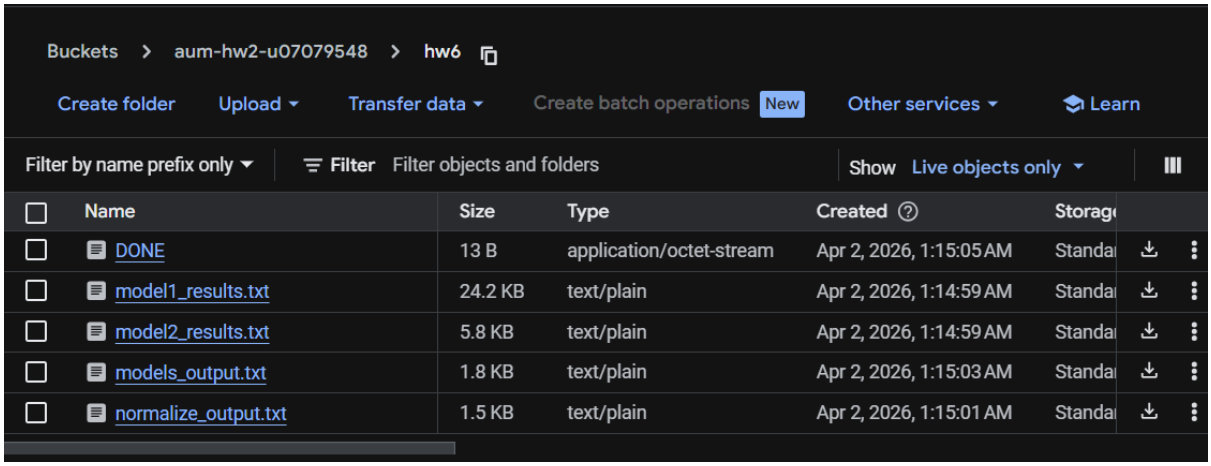
- Algorithm: RandomForestClassifier (100 trees, max_depth=20)
- Features: gender, age, time_of_day, country, is_banned (all label-encoded)
- Target: income bracket
- Train/test split: 80/20
- Income prediction is harder because the relationship between demographics and income in this synthetic data is noisy. The model aims for 40%+ accuracy.

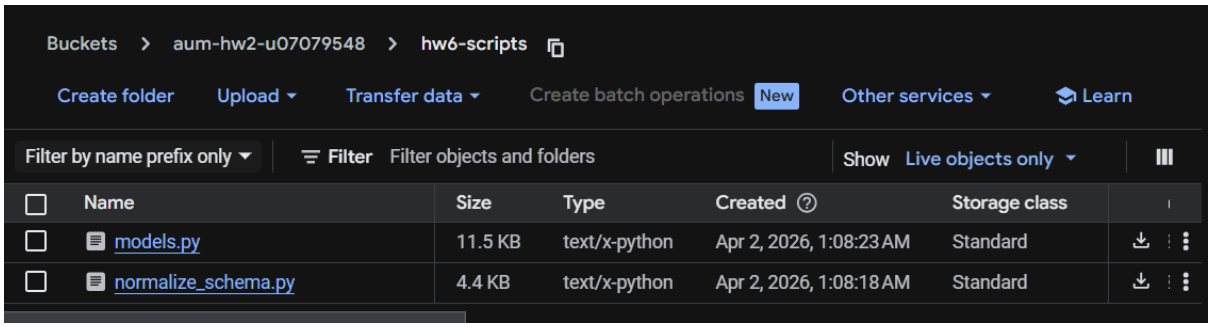
```
============================================================
MODEL 2: Predict Income from Request Fields
============================================================
  Samples with income: 100166
  Features       : gender, age, time_of_day, country, is_banned
  Target         : income
  Unique incomes : 8
  Income classes : [np.str_('0-10k'), np.str_('100k-150k'), np.str_('10k-20k'), np.str_('150k-250k'), np.str_('20k-40k
'), np.str_('250k+'), np.str_('40k-60k'), np.str_('60k-100k')]
  Train size     : 80132
  Test size      : 20034
  Training time  : 6.81s
  Test accuracy  : 64.43%

  Results written to /tmp/hw6_output/model2_results.txt
```

Step 4) - Test Set Output + GCS

- Both models' test set predictions were written as separate files to the GCS bucket.





Step 5) - Setup & Cleanup Scripts

- "The setup.sh script performs all operations in a single command: starts the database, creates a VM, runs both models, prints the output, deletes the VM, and stops the database."
- How to run: cd hw6/ && bash setup.sh

```
Aum@DESKTOP-1KD3K3T MINGW64 ~
$ cd "d:/Masters Studies/CS 528 - Cloud Computing/HW/Github Repo/AUM-U07079548-CLOUD-COMPUTING-HW/hw6"
bash setup.sh
=== HW6 Setup ===
Project : u0709548-aum-hw1
Region  : us-central1 / us-central1-a
[1/9] Enabling APIs ...
Operation "operations/acat.p2-1097067927322-8360cfe9-da0f-4e04-82f1-2255d4998d23" finished successfully.


Updates are available for some Google Cloud CLI components.  To install them,
please run:
  $ gcloud components update

[2/9] Uploading HW6 scripts to GCS ...
Copying file://normalize_schema.py to gs://aum-hw2-u07079548/hw6-scripts/normalize_schema.py
  Completed files 1/1 | 4.3kiB/4.3kiB
Copying file://models.py to gs://aum-hw2-u07079548/hw6-scripts/models.py
  Completed files 1/1 | 11.2kiB/11.2kiB
[3/9] Starting Cloud SQL instance 'hw5-mysql' ...
  Instance is STOPPED — starting ...
The following message will be used for the patch API method.
{"name": "hw5-mysql", "project": "u0709548-aum-hw1", "settings": {"activationPolicy": "ALWAYS"}}
Patching Cloud SQL instance...done.
Updated [https://sqladmin.googleapis.com/sql/v1beta4/projects/u0709548-aum-hw1/instances/hw5-mysql].
  Waiting for instance to become RUNNABLE ...
    state=RUNNABLE (attempt 1/30)
  DB_CONN_NAME=u0709548-aum-hw1:us-central1:hw5-mysql
[4/9] Creating service account and IAM bindings ...
Updated IAM policy for project [u0709548-aum-hw1].
bindings:
- members:
  - serviceAccount:service-1097067927322@gcp-sa-artifactregistry.iam.gserviceaccount.com
  role: roles/artifactregistry.serviceAgent
- members:
  - serviceAccount:1097067927322@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcf-admin-robot.iam.gserviceaccount.com
  role: roles/cloudfunctions.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudscheduler.iam.gserviceaccount.com
  role: roles/cloudscheduler.serviceAgent
- members:
  - serviceAccount:hw6-ml-sa@u0709548-aum-hw1.iam.gserviceaccount.com
```

```
[6/9] Creating ML VM 'hw6-ml-vm' ...
Created [https://www.googleapis.com/compute/v1/projects/u0709548-aum-hw1/zones/us-central1-a/instances/hw6-ml-vm].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce
 the risk of widespread service disruption, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs
/networking/zonal-dns

NAME       ZONE           MACHINE_TYPE   PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP    STATUS
hw6-ml-vm  us-central1-a  e2-standard-2               10.128.0.9   34.57.238.251  RUNNING
  VM created. Startup script will install dependencies and run models.
[7/9] Waiting for models to complete (polling GCS for DONE marker) ...
  This may take 5-10 minutes (installing packages + training) ...
    Not done yet (attempt 1/60), waiting 30s ...
    Not done yet (attempt 2/60), waiting 30s ...
    Not done yet (attempt 3/60), waiting 30s ...
    Not done yet (attempt 4/60), waiting 30s ...
  Models completed! (attempt 5)
```

```
[9/9] Cleaning up ...
  Deleting VM 'hw6-ml-vm' ...
  Stopping Cloud SQL instance 'hw5-mysql' ...
bindings:
- members:
  - serviceAccount:service-1097067927322@gcp-sa-artifactregistry.iam.gserviceaccount.com
  role: roles/artifactregistry.serviceAgent
- members:
  - serviceAccount:1097067927322@cloudbuild.gserviceaccount.com
  role: roles/cloudbuild.builds.builder
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudbuild.iam.gserviceaccount.com
  role: roles/cloudbuild.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcf-admin-robot.iam.gserviceaccount.com
  role: roles/cloudfunctions.serviceAgent
- members:
  - serviceAccount:service-1097067927322@gcp-sa-cloudscheduler.iam.gserviceaccount.com
  role: roles/cloudscheduler.serviceAgent
- members:
  - serviceAccount:1097067927322@cloudservices.gserviceaccount.com
  role: roles/compute.instanceGroupManagerServiceAgent
- members:
  - serviceAccount:service-1097067927322@compute-system.iam.gserviceaccount.com
  role: roles/compute.serviceAgent
- members:
  - serviceAccount:service-1097067927322@containerregistry.iam.gserviceaccount.com
  role: roles/containerregistry.ServiceAgent
- members:
  - serviceAccount:1097067927322-compute@developer.gserviceaccount.com
  - serviceAccount:1097067927322@cloudservices.gserviceaccount.com
  - serviceAccount:u0709548-aum-hw1@appspot.gserviceaccount.com
  role: roles/editor
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/logging.logWriter
- members:
  - user:aum1612@bu.edu
  role: roles/owner
- members:
  - serviceAccount:hw3-service-668@u0709548-aum-hw1.iam.gserviceaccount.com
  role: roles/pubsub.publisher
- members:
  - serviceAccount:service-1097067927322@gcp-sa-pubsub.iam.gserviceaccount.com
  role: roles/pubsub.serviceAgent
```